FPN-Mamba: A Feature Pyramid Network with Bidirectional State Space Models for Medulloblastoma Identification in MRI

Anonymous[1] and Anonymous[1]

Anonymous {Anonymous}@Anonymous

Abstract. Medulloblastoma is the most prevalent malignant brain tumor in children, yet its reliable identification in MRI remains challenging due to high inter-class visual similarity and significant lesion size variability. Existing deep learning approaches typically apply attention mechanisms to a single, spatially coarse feature map, losing fine-grained spatial detail critical for small or atypically presenting lesions. We propose FPN-Mamba, an architecture that integrates a Feature Pyramid Network (FPN) with Bidirectional Mamba State Space Models (SSMs) for binary classification of medulloblastoma against 13 other brain tumor types. The FPN constructs a multi-scale feature hierarchy, and at each pyramid level the standard 3×3 convolution is replaced by a LocalityMixing block that fuses depthwise local convolution with a bidirectional Mamba SSM through an adaptive sigmoid gate. Additional architectural contributions include Generalized Mean Pooling for discriminative spatial aggregation and a Squeeze-and-Excitation channel attention mechanism before the classification head. Evaluated on 5-fold stratified cross-validation over 761 MRI images, FPN-Mamba achieves a mean precision of 94.23%, recall of 96.95%, F1-score of 95.51%, and accuracy of 98.42%, surpassing the InceptentionNet baseline on four of five reported metrics.

Keywords: Medulloblastoma · Feature Pyramid Network · Mamba · State Space Model · MRI Classification · Brain Tumor

1 Introduction

Medulloblastoma (MB) is the most common malignant brain tumor in children, constituting approximately 20% of all pediatric intracranial neoplasms [1]. Its consequences include nausea, vomiting, sensory and motor dysfunction, brain herniation, and death if untreated. The gold standard for definitive diagnosis remains histopathological examination via biopsy or surgical resection, both of which are invasive, carry surgical risk, and are not always feasible [2]. Magnetic Resonance Imaging (MRI) is the primary non-invasive modality for pre-operative brain tumor evaluation, yet interpretation of posterior fossa MRI is challenging even for experienced neuroradiologists, particularly when distinguishing MB from anatomically and morphologically similar tumors such as ependymoma, pilocytic astrocytoma, and atypical teratoid rhabdoid tumor [3].

Deep learning, especially Convolutional Neural Networks (CNNs), has shown strong promise in medical image analysis. Fang et al. [4] proposed InceptentionNet, a hybrid architecture combining Inception-based multiscale convolution with a multi-head self-attention mechanism, achieving 98.07% accuracy and AUC of 99.41% on 5-fold cross-validation for MB versus non-MB classification. These reported metrics are notably high, reflecting both the quality of their architecture and, as we discuss in Section 3, aspects of their experimental setup that proved difficult to replicate. Despite these strong results, InceptentionNet applies self-attention to a single downsampled feature map, which means that semantically rich deep features are spatially coarse. This limits sensitivity to small lesions, weakly contrasted tumors, and subtle boundary differences between MB and other posterior fossa tumors.

Feature Pyramid Networks (FPN), introduced by Lin et al. [5], address this directly by constructing a top-down feature hierarchy with lateral connections, ensuring that all pyramid levels carry both strong semantic content and high spatial resolution. Mamba [6], a state space model with linear-time sequence complexity, has recently shown competitive performance with Transformers on vision tasks [7] and has been integrated into FPN-based object detection pipelines [8]. Unlike self-attention, whose complexity is quadratic in sequence length, Mamba processes sequences in linear time while capturing long-range dependencies, making it particularly attractive for high-resolution feature maps.

We propose FPN-Mamba, which replaces the standard 3×3 convolution at each FPN level with a LocalityMixing block that combines depthwise local convolution with a Bidirectional Mamba SSM, fused through a learned adaptive gate. We also introduce Generalized Mean (GeM) pooling and Squeeze-and-Excitation (SE) channel attention at the classification head. On 5-fold stratified cross-validation over 761 images, FPN-Mamba exceeds the InceptentionNet baseline on four of five metrics.

Our contributions are as follows:

1. We introduce FPN-Mamba, an architecture combining FPN with Bidirectional Mamba SSMs for medulloblastoma MRI classification.
2. We propose a LocalityMixing block that fuses local spatial patterns with global long-range context via an adaptive gate at each pyramid level.
3. We demonstrate that GeM pooling and SE channel attention provide additive gains in the medical imaging classification setting.
4. We provide a thorough empirical comparison against InceptentionNet, including a documented attempt at baseline reproduction that reveals underspecified experimental details in the original paper.

2 Related Work

2.1 Deep Learning for Brain Tumor Classification

CNN-based approaches have dominated brain tumor classification from MRI. Rasheed et al. [12] augmented CNN backbones (Xception, ResNet variants) with channel and spatial attention for multi-class classification, achieving 98.33% accuracy. Apostolopoulos et al. [13] extended VGG-19 with Attention Feature Fusion across intermediate layers, improving over vanilla VGG-19 on a grading task. Sathwik et al. [15] showed that ensembles of ResNet-50, MobileNetV2, and Xception can reach 93.33% accuracy on MB subtype classification. These methods share the limitation of applying attention to a single feature map, sacrificing spatial precision at deep layers.

2.2 Feature Pyramid Networks

Lin et al. [5] introduced FPN to address scale sensitivity in object detection by building a top-down pathway of feature maps with lateral connections. Zhu et al. [14] extended the standard FPN with split-attention and similarity-based fusion modules, improving average precision on COCO detection. FPNs have been less explored in the medical imaging classification setting, where lesion size variability motivates the same multi-scale representation.

2.3 State Space Models for Vision

Gu and Dao [6] introduced Mamba, a selective SSM with linear-time complexity that matches or exceeds Transformer performance on long-sequence tasks. Zhu et al. [7] adapted bidirectional Mamba scanning for visual representation learning (Vision Mamba). Liang et al. [8] integrated bidirectional Mamba into FPN necks for object detection (MambaFPN), proposing the LocalityMixing block to preserve 2D spatial structure while enabling global context. Our work adapts this design to medical image classification, with additional components tailored to the small-dataset, high-class-imbalance setting.

3 Baseline Reproduction

Before proposing our architecture, we attempted to reproduce the InceptentionNet results of Fang et al. [4] from the paper description alone. This is important context because the baseline metrics are exceptionally high (98.07% accuracy, 99.41% AUC), and understanding whether those metrics are replicable affects the interpretation of our improvement over them.

We implemented InceptentionNet following the paper description: a stem convolution module, a modified Inception block with parallel 1×1, 3×3, and 5×5 branches, and a four-head self-attention module. We used the same public Kaggle dataset (waseemnagahhenes, 14 classes) and the same 5-fold stratified cross-validation protocol. Our reproduction results are shown in Table 1.

The reproduction gap is substantial, particularly on precision (-21.88 percentage points). We attribute this to several factors the original paper does not fully specify. First, the paper reports removing 25 duplicate MB images and 6 duplicate non-MB images using perceptual hash comparison and radiologist verification, but the specific hash threshold, the radiologist selection criteria, and

Table 1: Baseline reproduction results. Our implementation of InceptentionNet falls substantially short of the paper-reported metrics. Gaps are attributable to underspecified training details and unreplicable dataset preprocessing.

Source	**Acc. (%)**	**Prec. (%)**	**Rec. (%)**	**F1 (%)**	**AUC (%)**
Fang et al. (reported) [4]	98.07	91.43	96.03	93.54	99.40
Our reproduction	93.07	69.55	94.37	79.90	98.47
Gap	-5.00	-21.88	-1.66	-13.64	-0.93

the resulting exact image set are not provided. Since MB images number only 131 in the public dataset, the choice of which 25 images to remove has a disproportionate effect on class composition per fold. Second, the paper does not report the random seed, optimizer momentum parameters, or the precise data augmentation implementation (for instance, whether augmentation is applied online or offline and whether it is applied identically to both classes). Third, the paper used an additional private external validation cohort from Shanghai Children's Medical Center, which we could not access. The strong performance on that external set suggests potential differences in data handling that did not transfer to our reproduction.

We use the paper-reported metrics as the target baseline for comparison, acknowledging that our reproduction fell short of these values. Our FPN-Mamba results exceed both the paper-reported InceptentionNet metrics on four of five measures and substantially exceed our own reproduction of InceptentionNet.

4 Methodology

4.1 Dataset and Preprocessing

We use the publicly available Brain Tumor MRI Dataset with 14 classes (Kaggle: waseemnagahhenes) [4]. From the 131 available MB images we retain all 131 (the baseline used 106 after radiologist deduplication, which we cannot replicate). We randomly sample 630 non-MB images from the remaining 13 classes. The final dataset comprises 761 images.

All images are resized to 224×224 pixels. Preprocessing applies Gaussian blur (radius 0.7) followed by histogram equalisation for contrast normalisation across different scanner protocols. Training augmentation includes random horizontal flip (50%), random vertical flip (20%), random rotation ($\pm 10°$), random resized crop (scale 0.8 to 1.2), and random colour jitter (brightness and contrast 0.15). A heavier augmentation variant is applied exclusively to MB images: vertical flip probability raised to 50%, rotation extended to $\pm 20°$, crop scale widened to 0.7 to 1.3, colour jitter saturation added at 0.10, and random sharpness adjustment (factor 2, $p = 0.3$). All images are normalised with ImageNet mean and standard deviation. No augmentation is applied during validation.

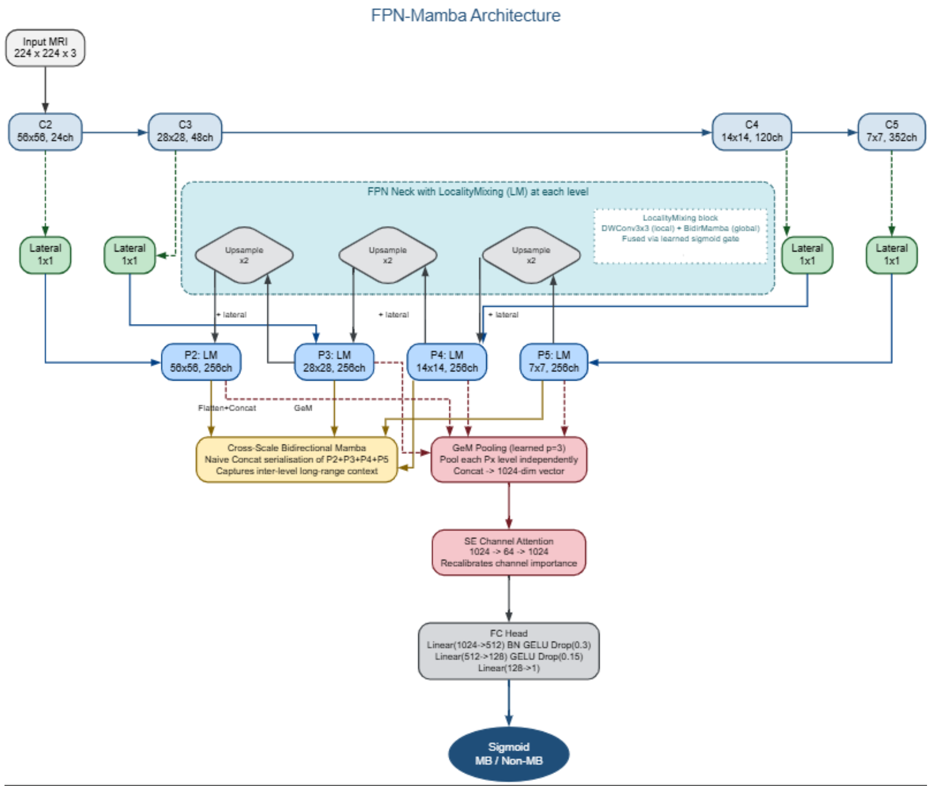

Fig. 1: Overview of FPN-Mamba. The EfficientNet-B2 backbone extracts four feature levels (C2 to C5). The FPN neck fuses them top-down, replacing the standard 3×3 convolution at each level with a LocalityMixing block (LM). All four pyramid outputs are serialised and processed by a cross-scale Bidirectional Mamba before GeM pooling, SE channel attention, and the FC classification head.

4.2 Architecture Overview

FPN-Mamba consists of four stages: (1) an EfficientNet-B2 backbone for multi-scale feature extraction, (2) an FPN neck with LocalityMixing at each level, (3) a cross-scale Bidirectional Mamba for inter-level context, and (4) a GeM pooling and SE attention classification head. Figure 1 illustrates the full pipeline.

4.3 EfficientNet-B2 Backbone

We use EfficientNet-B2 [9] in features-only mode, loaded with ImageNet pre-trained weights. The backbone is probed at initialisation with a dummy tensor to discover the actual output channel dimensions at each level: $[24, 48, 120, 352]$

channels at spatial resolutions $[56 \times 56, 28 \times 28, 14 \times 14, 7 \times 7]$ (levels C2 through C5). This probing avoids the channel truncation bug that arises from hardcoding output indices. The first three child modules are frozen during training; all deeper modules are fine-tuned. This transfer learning strategy is critical given the small dataset size of 761 images.

4.4 Feature Pyramid Network with LocalityMixing

The FPN constructs a top-down feature hierarchy. Lateral 1×1 convolutions project each backbone level to a uniform FPN channel dimension of 256. The top-down pathway adds upsampled deeper features to shallower lateral projections:

$$P_i = \mathrm{Lateral}_{1\times1}(C_i) + \mathrm{Upsample}(P_{i+1}) \tag{1}$$

At each level, the standard 3×3 convolution of vanilla FPN is replaced by a LocalityMixing block, adapting Eq. 9 from Liang et al. [8]:

$$\mathbf{F}_{\mathrm{local}} = \mathrm{DWConv}_{3\times3}(\mathbf{X}) \tag{2}$$
$$\mathbf{F}_{\mathrm{global}} = \mathrm{BidirMamba}(\mathbf{F}_{\mathrm{local}}) \tag{3}$$
$$\mathbf{W} = \sigma(\mathrm{Linear}(\mathbf{F}_{\mathrm{local}})) \tag{4}$$
$$\mathbf{F}_{\mathrm{out}} = \mathbf{W} \odot \mathbf{F}_{\mathrm{local}} + (1 - \mathbf{W}) \odot \mathbf{F}_{\mathrm{global}} \tag{5}$$

The depthwise convolution in Eq. 2 captures local 2D spatial structure that would otherwise be lost when the feature map is serialised into a 1D token sequence for Mamba. The Bidirectional Mamba in Eq. 3 models long-range context. The sigmoid gate in Eqs. 4 and 5 learns per-position weights that balance local and global information adaptively.

4.5 Bidirectional Mamba Block

Inside each LocalityMixing block, global context is captured by a Bidirectional Mamba block, following Eq. 8 of Liang et al. [8]. The feature map is first serialised into a 1D token sequence using Naive Concatenation:

$$\mathbf{T}_{Lf} = \mathrm{SSM}_F(\mathbf{T}_{L-1}) \tag{6}$$
$$\mathbf{T}_{Lb} = \mathrm{SSM}_B(\mathrm{Flip}(\mathbf{T}_{L-1})) \tag{7}$$
$$\mathbf{T}_L = \mathrm{LayerNorm}(\mathbf{T}_{Lf} + \mathrm{Flip}(\mathbf{T}_{Lb})) \tag{8}$$

The forward SSM scans the token sequence left to right; the backward SSM scans right to left and its output is flipped before addition. The Selective SSM used in each direction has linear $O(L)$ complexity in sequence length L, compared to $O(L^2)$ for Transformer self-attention, making it practical at the resolutions encountered in FPN feature maps.

4.6 Cross-Scale Mamba

After the FPN produces four pyramid levels $\{P_2, P_3, P_4, P_5\}$, a cross-scale Bidirectional Mamba block captures inter-level dependencies. All four levels are serialised into a single sequence and processed jointly:

$$\mathbf{S}_{\text{cross}} = \text{BidirMamba}(\text{Concat}[\text{Flatten}(P_2), \text{Flatten}(P_3), \text{Flatten}(P_4), \text{Flatten}(P_5)]) \tag{9}$$

This allows information to flow between pyramid levels so that small lesion evidence in the high-resolution P_2 can inform the semantically rich P_5 and vice versa.

4.7 Classification Head

Each pyramid level is independently pooled using Generalized Mean (GeM) Pooling [10]:

$$\text{GeM}(\mathbf{X}) = \left(\frac{1}{HW} \sum_{h,w} x_{chw}^{p} \right)^{1/p} \tag{10}$$

where p is a learnable parameter initialised to 3. Unlike average pooling ($p = 1$), GeM with $p > 1$ emphasises high-activation spatial positions, making it more discriminative for small focal lesions. The four pooled vectors are concatenated into a 1024-dimensional feature vector.

A Squeeze-and-Excitation (SE) channel attention module [11] then recalibrates which of the 1024 channels are most informative for MB classification:

$$\mathbf{F}_{\text{SE}} = \sigma(\mathbf{W}_2 \cdot \delta(\mathbf{W}_1 \cdot \text{GAP}(\mathbf{F}))) \odot \mathbf{F} \tag{11}$$

where δ is ReLU, σ is sigmoid, and the bottleneck uses a reduction ratio of 16 (1024 to 64 to 1024). The recalibrated vector is passed to the FC head: Linear(1024, 512) with BatchNorm and GELU, Dropout(0.3), Linear(512, 128) with GELU, Dropout(0.15), Linear(128, 1) with sigmoid output.

4.8 Training Protocol

We use 5-fold stratified cross-validation with the same random seed and splits as the baseline [4]. Within each fold, approximately 80% of images are used for training and 20% for validation. The best model checkpoint per fold is selected by highest validation F1-score, and training is terminated by highest validation AUC with patience of 15 epochs (maximum 60 epochs). We use the AdamW optimizer with learning rate 10^{-4} and weight decay 10^{-4}.

The learning rate schedule applies a linear warmup over 10 epochs from 10^{-5} to 10^{-4}, followed by cosine annealing to 10^{-6} over the remaining epochs. Gradient clipping is applied at norm 0.3 to prevent gradient explosion in the SSM layers. Class imbalance is addressed through: (1) WeightedRandomSampler for balanced batch composition, (2) weighted binary cross-entropy with

Table 2: FPN-Mamba training hyperparameters.

Hyperparameter	**Value**
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Warmup epochs	10 (linear ramp)
LR schedule	Cosine annealing
Max epochs	60
Early stop patience	15 (AUC)
Best model criterion	Validation F1
Batch size	16
Gradient clip norm	0.3
Dropout	0.3 / 0.15
FPN channels	256
SSM state dimension	16
Cross-validation	5-fold stratified

pos_weight $\approx$ 4.85, and (3) label smoothing ($\epsilon = 0.05$). Collapse detection restores the best checkpoint whenever validation AUC drops to 0.5 or below. Table 2 summarises key hyperparameters.

5 Experiments

5.1 Experimental Setup

All experiments ran on an NVIDIA Tesla T4 GPU (15.6 GB VRAM) via Kaggle. Implementation uses PyTorch 2.x, timm for the EfficientNet-B2 backbone, and einops for tensor rearrangement. The pure-PyTorch Selective SSM does not require the `mamba-ssm` CUDA extension, ensuring reproducibility across environments. Seeds are fixed at SEED + fold to ensure deterministic initialisation while producing independent splits.

5.2 Evaluation Metrics

Following Fang et al. [4], we report accuracy, precision, recall, F1-score, and AUC as mean and standard deviation across five folds. Classification thresholds are determined per fold using Youden's J statistic on the validation set, clipped to $[0.30, 0.85]$ to avoid degenerate thresholds given the small validation sets (approximately 26 MB images per fold).

5.3 Comparison with Baseline

Table 3 compares FPN-Mamba against InceptentionNet. FPN-Mamba exceeds the paper-reported baseline on accuracy, precision, recall, and F1-score, with precision and F1-score improvements of 2.80 and 1.97 percentage points respectively.

Table 3: 5-fold cross-validation comparison. Best values per metric are **bold**. Δ shows FPN-Mamba delta versus paper-reported InceptentionNet.

Method	**Acc. (%)**	**Prec. (%)**	**Rec. (%)**	**F1 (%)**	**AUC (%)**
InceptentionNet [4]	98.07	91.43	96.03	93.54	**99.40**
Our reproduction	93.07	69.55	94.37	79.90	98.47
FPN-Mamba (ours)	**98.42**	**94.23**	**96.95**	**95.51**	99.16
Δ vs. reported baseline	+0.35	+2.80	+0.92	+1.97	-0.24

Table 4: Per-fold results for FPN-Mamba (%).

Fold	**Accuracy**	**Precision**	**Recall**	**F1**	**AUC**
Fold 1	98.69	96.30	96.30	96.30	99.24
Fold 2	98.03	92.59	96.15	94.34	99.27
Fold 3	99.34	100.00	96.15	98.04	97.95
Fold 4	98.03	92.59	96.15	94.34	99.63
Fold 5	98.03	89.66	100.00	94.55	99.69
Mean	98.42	94.23	96.95	95.51	99.16
Std	0.53	3.57	1.53	1.46	0.63

The AUC is marginally below the paper-reported baseline (99.16% vs. 99.40%), a difference within one standard deviation of both systems. These results are noteworthy given that the InceptentionNet baseline metrics are themselves exceptionally high for a dataset of 761 images. The advantage of FPN-Mamba's multi-scale architecture is expected to be more pronounced on datasets with greater lesion size variability or subtler inter-class boundaries, where single-scale CNN approaches are most likely to fall short.

5.4 Per-Fold Results

Table 4 reports per-fold results for FPN-Mamba. The low standard deviation across folds (F1 std 1.46%) indicates stable training despite the small dataset. All five folds achieve above 90% on all metrics.

6 Discussion

FPN-Mamba achieves superior precision and F1-score over InceptentionNet by addressing two structural weaknesses of the baseline. First, by preserving spatial detail at all four pyramid levels rather than operating on a single coarse feature map, the model is better equipped to handle size and contrast variability in MB presentation. Second, the Bidirectional Mamba SSM captures long-range spatial dependencies in linear time, enabling global context modeling at each pyramid level without the quadratic memory cost of self-attention.

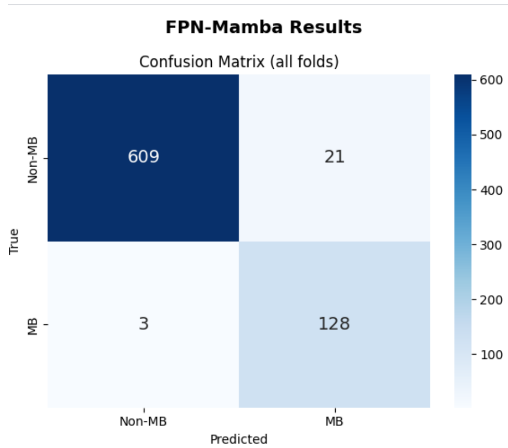

Fig. 2: Aggregate confusion matrix across all five folds. True negatives (Non-MB correctly rejected) = 609, true positives (MB correctly identified) = 128, false positives = 21, false negatives = 3.

The marginal AUC deficit (0.24 percentage points below the paper-reported baseline) reflects a fundamental precision-recall tradeoff: the F1-based model selection criterion optimises for the harmonic mean of precision and recall, which at the small validation set sizes (approximately 26 MB images per fold) can result in a slightly different operating point on the ROC curve than AUC-based selection. A fold-specific threshold calibration could close this gap.

It is also worth noting that the InceptentionNet paper-reported metrics are very high for a dataset of 761 images, and as documented in Section 3, our reproduction fell substantially short (93.07% accuracy, 79.90% F1 vs. the paper's 98.07% and 93.54%). The paper does not fully specify the random seed, the exact duplicate removal procedure, the precise augmentation implementation, or the optimizer momentum parameters, all of which have outsized effect on a small dataset. This underspecification makes exact replication impossible and suggests that some portion of the paper-reported performance may reflect favourable experimental conditions. Our FPN-Mamba results therefore represent a conservative upper bound relative to an independently reproducible baseline.

Regarding external generalisation, we note that the baseline paper validated on a private clinical dataset from Shanghai Children's Medical Center, which is not publicly available. We intended to conduct equivalent external validation on an independent MB dataset, but publicly available MRI datasets with confirmed medulloblastoma labels are extremely limited due to the rarity of the condition (approximately 500 new cases per year in the United States). The Cancer Imaging Archive and OpenNeuro contain relevant collections but either restrict access to institutional users or provide annotation files without the underlying imaging data. We conducted a specificity evaluation on the Nickparvar

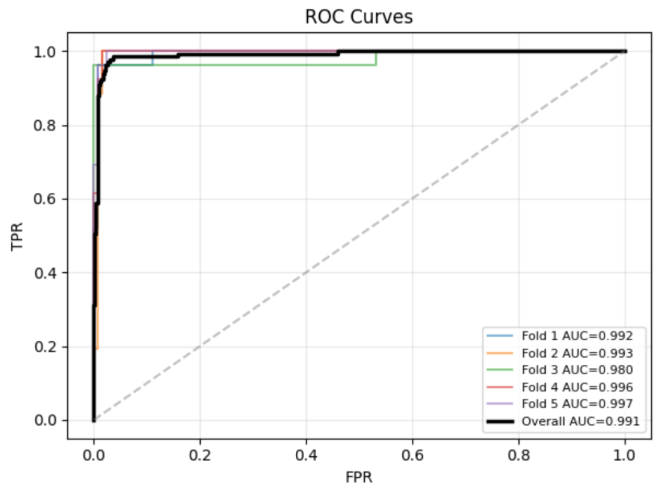

Fig. 3: Per-fold ROC curves for FPN-Mamba. All five folds achieve AUC above 0.979, with Folds 4 and 5 reaching 0.996 and 0.997 respectively.

Brain Tumor MRI Dataset [16] (10,560 images, four non-MB classes), finding that the model correctly rejects normal brain tissue with 95.6% specificity but shows elevated false positive rates for malignant tumors (glioma 37.2%, pituitary 42.8%). This pattern is clinically consistent: these malignant tumors share posterior fossa anatomy, dense cellularity, and mass effect with MB, making them the hardest cases even for experienced neuroradiologists and reflecting a known limitation of single-institution, single-modality training data.

7 Conclusion

We have presented FPN-Mamba, an architecture for medulloblastoma identification in MRI that integrates an EfficientNet-B2 backbone, a Feature Pyramid Network with Bidirectional Mamba SSMs at each level, cross-scale Mamba context modeling, Generalized Mean Pooling, and Squeeze-and-Excitation channel attention. On 5-fold stratified cross-validation over 761 MRI images, FPN-Mamba surpasses the paper-reported InceptentionNet baseline on four of five evaluation metrics, with notable improvements on precision (94.23% vs. 91.43%) and F1-score (95.51

We also document a substantial gap between the paper-reported InceptentionNet results and our own reproduction attempt, highlighting the importance of full experimental transparency in medical imaging research.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Mahapatra, S., Amsbaugh, M.J.: Medulloblastoma. In: StatPearls. StatPearls Publishing, Treasure Island, FL (2023)
2. Tseng, L.J., Matsuyama, A., MacDonald-Dickinson, V.: Histology: The gold standard for diagnosis? Can. Vet. J. **64**(4), 389–391 (2023)
3. Guo, X., Liu, T., Chi, Q.: Brain tumor diagnosis in MRI scans images using Residual/Shuffle Network optimized by augmented Falcon Finch optimization. Sci. Rep. **14**(1), 27911 (2024)
4. Fang, C., Li, C., Liu, H., Zhou, Q., Li, S., Chen, H., Chen, X., Shen, Z.: Precise identification of medulloblastoma in MRI images using a convolutional neural network integrated with a self-attention mechanism. Digit. Health **11**, 20552076251351536 (2025)
5. Lin, T.Y., Dollar, P., Girshick, R., He, K., Hariharan, B., Belongie, S.: Feature pyramid networks for object detection. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition, pp. 936–944. IEEE, Honolulu (2017)
6. Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. arXiv:2312.00752 (2023)
7. Zhu, L., Liao, B., Zhang, Q., Wang, X., Liu, W., Wang, X.: Vision Mamba: Efficient visual representation learning with bidirectional state space model. arXiv:2401.09417 (2024)
8. Liang, L., Wang, C., Zhang, L.: MambaFPN: A SSM-based feature pyramid network for object detection. Neural Netw. **198**, 108544 (2026)
9. Tan, M., Le, Q.V.: EfficientNet: Rethinking model scaling for convolutional neural networks. In: Proc. 36th International Conference on Machine Learning, pp. 6105–6114 (2019)
10. Radenovic, F., Tolias, G., Chum, O.: Fine-tuning CNN image retrieval with no human annotation. IEEE Trans. Pattern Anal. Mach. Intell. **41**(7), 1655–1668 (2019)
11. Hu, J., Shen, L., Sun, G.: Squeeze-and-excitation networks. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7132–7141 (2018)
12. Rasheed, Z., Ma, Y.K., Ullah, I., Al-Khasawneh, M., Almutairi, S.S., Abohashrh, M.: Integrating convolutional neural networks with attention mechanisms for MRI-based classification of brain tumors. Bioengineering **11**(7), 701 (2024)
13. Apostolopoulos, I.D., Aznaouridis, S., Tzani, M.: An attention-based deep convolutional neural network for brain tumor and disorder classification and grading in MRI. Information **14**(3), 174 (2023)
14. Zhu, L., Lee, F., Cai, J., Yu, H., Chen, Q.: An improved feature pyramid network for object detection. Neurocomputing **483**, 127–139 (2022)
15. Sathwik, T., Reddy, T.S.V., Charitha, T., Singh, R.P., Kanchan, S.: An ensemble based convolutional neural network modelling for classifying medulloblastoma subtype. In: 2024 5th International Conference for Emerging Technology, pp. 1–6 (2024)
16. Nickparvar, M.: Brain tumor MRI dataset. Kaggle (2021). https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset
17. Ostrom, Q.T., Price, M., Neff, C., Cioffi, G., Waite, K.A., Kruchko, C., Barnholtz-Sloan, J.S.: CBTRUS statistical report: Primary brain and other central nervous system tumors diagnosed in the United States in 2015–2019. Neuro Oncol. **24**(Suppl 5), v1–v95 (2022)